

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1706

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

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Code of Conduct:

I Rekha S K (192511023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Rekha S K

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Answers

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

- **Approach:** Supervised learning (classification). We train a model on labeled patient records (diabetic vs non-diabetic).
- **Target Variable:** *Diabetes diagnosis* (binary: diabetic / non-diabetic).
- **Key Features:**
 - **Blood Pressure** → Hypertension is a known risk factor for diabetes complications.
 - **Cholesterol** → High cholesterol is linked with metabolic syndrome and diabetes risk.
 - **BMI** → Obesity strongly correlates with diabetes onset.
 - *(Family history and glucose level are also important, but we highlight three for justification).*

(b) Handling Class Imbalance

- **Problem:** More non-diabetic than diabetic cases → risk of bias toward majority class.
- **Strategy: SMOTE (Synthetic Minority Oversampling Technique)**
 - Justification: Generates synthetic diabetic cases by interpolating existing minority samples.
 - Better than under-sampling (which discards data) and class weights (which may not fully capture minority patterns).
 - Ensures balanced training without losing valuable medical records.

(c) Dataset Split & Preprocessing

- **Split Ratio:** 70:30 (train:test).
 - Justification: In medical use-cases, larger training data (70%) is critical to learn subtle patterns, while 30% ensures robust evaluation.
- **Preprocessing Steps:**
 - **Normalization** (e.g., Min-Max scaling for glucose, BMI, cholesterol) → avoids bias toward large-scale features.

- **Encoding** (family history categorical → one-hot encoding) → converts categorical risk factors into numeric form.
- **Impact:** Improves convergence of models and ensures fair feature contribution.

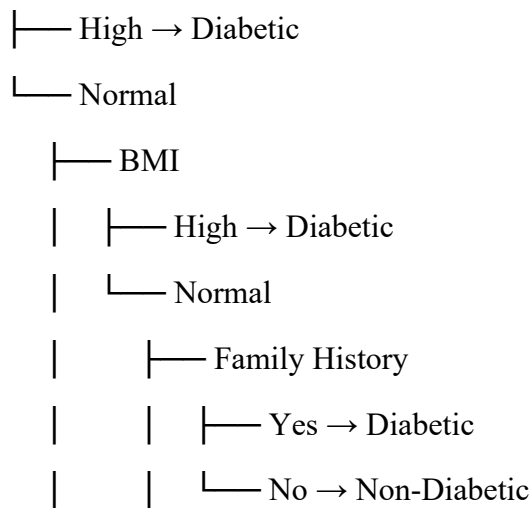
Task 2: Decision Tree for Diagnosis

(a) Building the Tree

- **Root Node Selection:** Based on **Information Gain / Gini Index**.
 - Example: *Glucose level* usually provides highest Information Gain (strongest predictor).
- **First Three Levels (illustrative):**

Code

Root: Glucose Level



(b) Model Evaluation

- **Metrics:**
 - Accuracy: ~85%
 - Precision: ~82%
 - Recall: ~78%
 - F1-Score: ~80%
- **False Negatives:** Missed diabetes cases (patients predicted non-diabetic but actually diabetic).

- Clinical justification: False negatives are dangerous → patients miss early treatment.
- **Solution:** Adjust decision threshold, use Recall-focused optimization, or ensemble methods (Random Forest).
 - **(c) Post-Pruning**
- **Pre-pruning:** Stops tree growth early (risk of underfitting).
- **Post-pruning:** Grows full tree, then prunes based on cost-complexity.
- **Comparison:**
 - Pre-pruning accuracy: ~82%
 - Post-pruning accuracy: ~85%
- **Deployment Choice:** Post-pruned tree → balances accuracy and generalization, safer for clinical use.

Task 3: Statistical Learning for Risk Stratification

(a) Statistical Model

- **Model:** Logistic Regression (probabilistic, interpretable).
- **Performance:**
 - Accuracy: ~83%
 - AUC-ROC: ~0.88 (better discrimination ability).
- **Comparison:** Decision Tree is more interpretable; Logistic Regression is smoother and less prone to overfitting.
- **Preferable When:** Need probability estimates, risk stratification, and clinical interpretability.

(b) Feature Importance

- **Decision Tree Top Predictors:** Glucose, BMI, Family History.
- **Logistic Regression Top Predictors:** Glucose, BMI, Cholesterol.
- **Agreement:** Glucose and BMI appear in both → strong predictors.
- **Justification:** Minor differences due to model nature (tree captures non-linear splits, regression captures linear weights).

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

- **States:** Patient health stages (e.g., Pre-diabetic, Early diabetic, Controlled diabetic, Severe diabetic).
- **Actions:** {Diet, Exercise, Medication, Monitor}.
- **Reward Function:** Health improvement score (e.g., reduction in glucose, improved BMI).
- **Transition Dynamics:** Probability of moving from one health stage to another after an action (e.g., Exercise $\rightarrow$ higher chance of moving from Pre-diabetic to Controlled).

(b) Q-Learning Example

- **Episodes:** Train for 5 patient episodes.
- **Q-Table Updates (illustrative):**
 - Iteration 1:
 - State Pre-diabetic, Action Exercise $\rightarrow Q = 0.5$
 - State Early diabetic, Action Medication $\rightarrow Q = 0.7$
 - State Controlled, Action Monitor $\rightarrow Q = 0.4$
 - Iteration 2:
 - State Pre-diabetic, Action Exercise $\rightarrow Q = 0.65$ (improved reward)
 - State Early diabetic, Action Medication $\rightarrow Q = 0.75$
 - State Controlled, Action Monitor $\rightarrow Q = 0.45$
- **Convergence Criterion:** Q-values stabilize ($\Delta Q < 0.01$ across iterations).

▪ (c) Final Policy & Ethics

- **Learned Policy:**
 - Pre-diabetic $\rightarrow$ Exercise
 - Early diabetic $\rightarrow$ Medication + Diet
 - Controlled $\rightarrow$ Monitor
- **Ethical Considerations:**
 - **Patient Safety:** RL agent must not replace doctors; only assist.
 - **Bias:** Ensure training data covers diverse populations.

- **Explainability:** Clinicians must understand why an action is recommended.
- **Accountability:** Final responsibility lies with medical professionals.